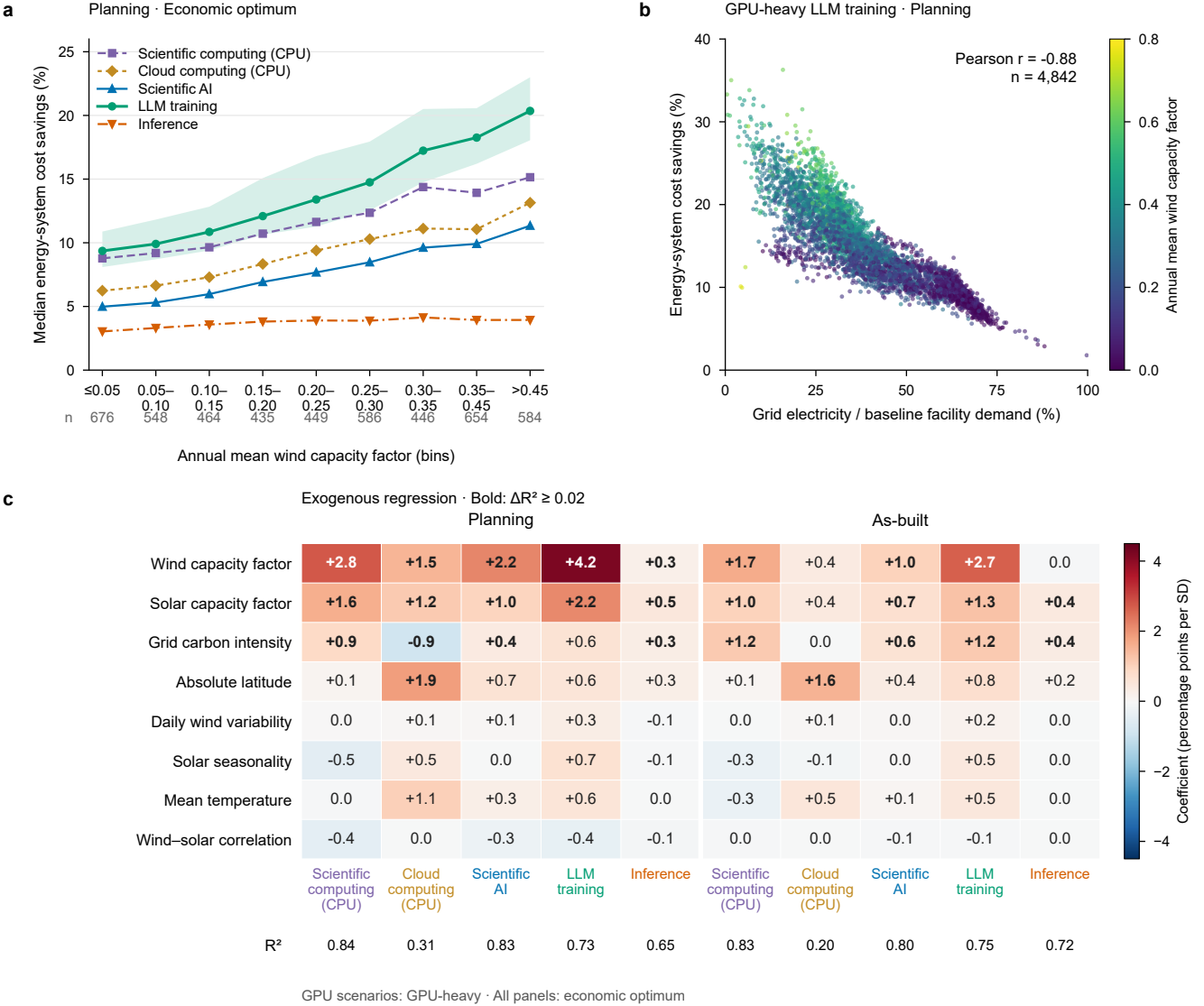

a
Planning · Economic optimum
Scientific computing (CPU)
Cloud computing (CPU)
Scientific AI
LLM training
Inference
Median energy-system cost savings (%)
25
20
15
10
5
0
≤0.05
0.05–0.10
0.10–0.15
0.15–0.20
0.20–0.25
0.25–0.30
0.30–0.35
0.35–0.45
>0.45
n 676 548 464 435 449 586 446 654 584
Annual mean wind capacity factor (bins)
b
GPU-heavy LLM training · Planning
Pearson r = -0.88
n = 4,842
Energy-system cost savings (%)
40
30
20
10
0
0
25
50
75
100
Grid electricity / baseline facility demand (%)
Annual mean wind capacity factor
0.8
0.6
0.4
0.2
0.0
c
Exogenous regression · Bold: ΔR² ≥ 0.02
Planning
As-built
Wind capacity factor +2.8 +1.5 +2.2 +4.2 +0.3 +1.7 +0.4 +1.0 +2.7 0.0
Solar capacity factor +1.6 +1.2 +1.0 +2.2 +0.5 +1.0 +0.4 +0.7 +1.3 +0.4
Grid carbon intensity +0.9 -0.9 +0.4 +0.6 +0.3 +1.2 0.0 +0.6 +1.2 +0.4
Absolute latitude +0.1 +1.9 +0.7 +0.6 +0.3 +0.1 +1.6 +0.4 +0.8 +0.2
Daily wind variability 0.0 +0.1 +0.1 +0.3 -0.1 0.0 +0.1 0.0 +0.2 0.0
Solar seasonality -0.5 +0.5 0.0 +0.7 -0.1 -0.3 -0.1 0.0 +0.5 0.0
Mean temperature 0.0 +1.1 +0.3 +0.6 0.0 -0.3 +0.5 +0.1 +0.5 0.0
Wind–solar correlation -0.4 0.0 -0.3 -0.4 -0.1 0.0 0.0 -0.1 -0.1 0.0
Scientific computing (CPU)
Cloud computing (CPU)
Scientific AI
LLM training
Inference
Scientific computing (CPU)
Cloud computing (CPU)
Scientific AI
LLM training
Inference
R² 0.84 0.31 0.83 0.73 0.65 0.83 0.20 0.80 0.75 0.72
Coefficient (percentage points per SD)
4
2
0
−2
−4
GPU scenarios: GPU-heavy · All panels: economic optimum